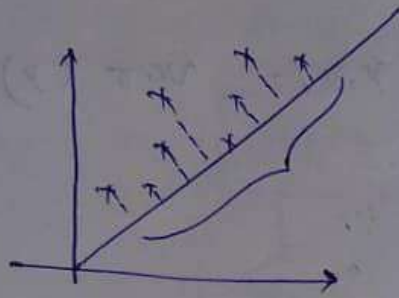


Eigen vector And Eigen values



[Linear Transformation]

[Eigen decomposition of covariance matrix]

↓
Eigen vector & Eigen values

$$\begin{bmatrix} \dots \\ \dots \end{bmatrix}$$

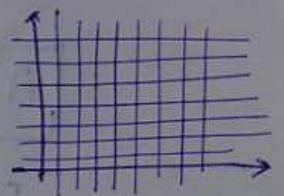
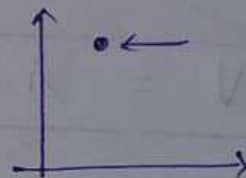
$$* [v] = \lambda * v$$

↓
Eigen value

$$A * V = \lambda * V$$

↑↑

↓↓



Eigen vector → Maximum magnitude

Eigen vector → Max magnitude Principal component

↓

Max Eigen vector Max var

↓

Best Principal Component → PC1

Steps to calculate Eigen value and vector

① Covariance of features

$$\begin{bmatrix} x, y \end{bmatrix}$$

z

↓
x'

$$cov(x, y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{N-1}$$

(x ₁ , y ₁)	(x ₂ , y ₂)
(x ₃ , y ₃)	(x ₄ , y ₄)

2x2

$$A = \begin{array}{c|cc} & x & y \\ \hline x & \text{Var}(x) & \text{Cov}(x,y) \\ \hline y & \text{Cov}(y,x) & \text{Var}(y) \end{array}$$

$$\text{Cov}(x,x) = \text{Var}(x)$$

$$\text{Cov}(y,y) = \text{Var}(y)$$

3x3

$$\begin{array}{c|ccc} & x & y & z \\ \hline x & & & \\ \hline y & & & \\ \hline z & & & \end{array}$$

$$\lambda_1, \lambda_2, \lambda_3$$

f_1 & f_2

$$A \cdot V = \lambda \cdot V$$

$$\lambda_1$$

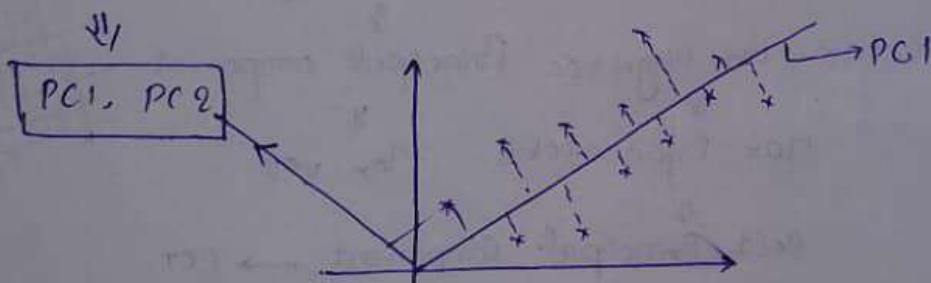
$$\lambda_2$$

↓
PC1

↓
PC2

$\lambda_1, \lambda_2 \rightarrow$ Eigen values

↓

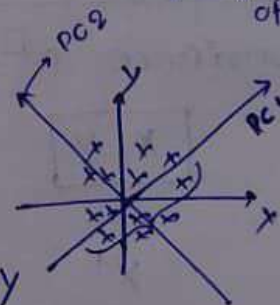


(*)



2D $\rightarrow$ 1D

$\lambda_1 \rightarrow$ magnitude of the Eigen vector



① standardize the data

② Covariance Matrix of x & y

$$\begin{array}{c|cc} & x & y \\ \hline x & \text{Var}(x) & \text{Cov}(x,y) \\ \hline y & \text{Cov}(y,x) & \text{Var}(y) \end{array}$$

③ Find out Eigen vector And value

$$A V = \lambda V$$

λ_1, λ_2 $\Rightarrow$ Eigen values



PC1



PC2

3D $\rightarrow$ 2D

$\lambda_1, \lambda_2, \lambda_3$



PC1



PC2



PC3



1D



1D = 2D

3D $\rightarrow$ 1D

$\lambda_1, \lambda_2, \lambda_3$



1D

2D $\rightarrow$ 1D

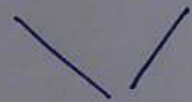
λ_1, λ_2



PC1



PC2



1D

Projection

